

Thomas Marconi

850-252-5128 | thomas.marconi@outlook.com | linkedin.com/in/thomasmarconi | github.com/thomasmarconi | marconi.bearblog.dev

EDUCATION

Georgia Institute of Technology

Atlanta, GA

M.S. Computer Science

Expected Dec 2027

- Coursework: Database System Implementation, High-Performance Computer Architecture, Computer Networks

Florida State University

Tallahassee, FL

B.S. Computer Science | GPA: 3.7

May 2023

EXPERIENCE

Tactical Air Support

Feb 2024 – Present

Software Engineer

Jacksonville, FL

Aerospace defense contractor supporting U.S. military operations

- Designed a real-time collaborative task management system with concurrent write coordination across distributed clients; solved event ordering and partial-failure recovery at the WebSocket layer to achieve consistent sub-100ms state propagation — without an external message broker or distributed lock.
- Replaced three vendor software systems with purpose-built internal tooling, eliminating **\$850K in annual licensing costs**; designed for high availability and zero-downtime deployments under the constraint of no dedicated operations team.
- Eliminated static credentials across a multi-service Azure deployment by architecting runtime secrets injection via managed identities and Key Vault; enforced least-privilege IAM boundaries per service principal to satisfy FedRAMP High audit controls.
- Built a retrieval pipeline over 5,000+ unstructured PDFs; designed overlapping semantic chunking to preserve context across boundaries and implemented two-stage retrieval — dense vector similarity followed by sparse reranking — to improve precision on narrow technical queries.
- Built a quality metrics pipeline in response to a formal Corrective Action Request; unified data from Smartsheet and QT9 into a FastAPI/MongoDB backend and exposed FPQR trends, CAR classification, and audit forecasting — converting a compliance requirement into a live operational dashboard.
- Designed a policy management system serving 500+ employees; implemented multi-stage approval workflows with async notification pipelines and real-time UI state in Next.js, replacing an untracked manual process with an auditable, observable governance system.

TECHNICAL PROJECTS

Project Desert | Next.js, Python (FastAPI), PostgreSQL, Docker

Jan 2025 – Present

- Enforced explicit API contracts via OpenAPI schema validation at service boundaries; auto-generated TypeScript clients from Pydantic models to surface type violations at compile time and eliminate a class of cross-service integration failures.
- Implemented stateless JWT authentication with server-side refresh token rotation across distributed services; engineered to handle concurrent refresh races and session invalidation without shared distributed state.

CityScale | React, FastAPI, PCA | 3rd Place — AI for Good Hackathon

Oct 2025

- Built a geospatial rendering pipeline capable of filtering and visualizing **580,000+ data points** at sub-second latency; implemented client-side spatial indexing and progressive data loading to maintain interactive frame rates at full dataset scale.
- Applied PCA to a high-dimensional economic dataset; selected retained components by explained variance ratio to produce a compact 2D city-level embedding while preserving the majority of discriminative signal.

TECHNICAL SKILLS

Languages: TypeScript, Python, SQL, JavaScript, C++, Java, Bash

Backend: Node.js, FastAPI, WebSockets, REST APIs, PostgreSQL, Docker, Podman

AI & Data: Vector Retrieval (ChromaDB, Azure Semantic Search), LLM Orchestration (LangGraph), Pandas, scikit-learn

Cloud & Infrastructure: Azure (Functions, Managed Identity, Key Vault, Entra ID), CI/CD, Linux

Frontend: React, Next.js, Zustand, TanStack Query, Tailwind CSS